

Day 16 – Shell Scripting Basics

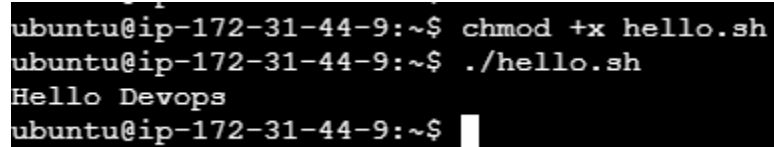
Task 1: Your First Script

shebang (`#!/bin/bash`) and why it matters- The shebang (`#!/bin/bash`) is a special line at the very beginning of a script file that indicates which interpreter should be used to execute the script.

[hello.sh](#)

```
#!/bin/bash
```

```
echo "Hello Devops"
```

A terminal window with a black background and white text. The prompt is 'ubuntu@ip-172-31-44-9:~\$'. The first command is 'chmod +x hello.sh'. The second command is './hello.sh'. The output is 'Hello Devops'. The prompt is now 'ubuntu@ip-172-31-44-9:~\$' with a cursor.

```
ubuntu@ip-172-31-44-9:~$ chmod +x hello.sh
ubuntu@ip-172-31-44-9:~$ ./hello.sh
Hello Devops
ubuntu@ip-172-31-44-9:~$
```

Task 2: Variables

[variables.sh](#)

```
#!/bin/bash
```

```
Name="Parthavi"
```

```
Role="Cloud Engineer"
```

```
echo "Hello, I am $Name and my role is $Role"
```

```

ubuntu@ip-172-31-44-9:~$ ls -l variables.sh
-rw-rw-r-- 1 ubuntu ubuntu 97 Jun  1 04:20 variables.sh
ubuntu@ip-172-31-44-9:~$ chmod +x variables.sh
ubuntu@ip-172-31-44-9:~$ ls -l
total 32
drwxrwxr-x 2 berlin    heist-team 4096 May 31 17:20 app-logs
drwxrwxr-x 2 ubuntu    ubuntu    4096 May 31 17:48 bank-heist
-r--r--r-- 1 berlin    ubuntu     0 May 31 14:46 devops.txt
drwxrwxr-x 4 professor planners  4096 May 31 17:30 heist-project
-rwxrwxr-x 1 ubuntu    ubuntu     33 Jun  1 04:15 hello.sh
-rw-r----- 1 ubuntu    ubuntu     12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu    ubuntu    4096 May 31 15:06 project
-rw-rw-r-- 1 professor heist-team  0 May 31 17:14 project-config.yml
-rw-rw-r-- 1 professor heist-team  0 May 31 17:18 project-config1.yml
-rw-rw-r-- 1 ubuntu    ubuntu     13 May 31 14:52 script.sh
-rw-rw-r-- 1 ubuntu    heist-team  0 May 31 17:09 team-notes.txt
-rwxrwxr-x 1 ubuntu    ubuntu     97 Jun  1 04:20 variables.sh
ubuntu@ip-172-31-44-9:~$ ls -l variables.sh
-rwxrwxr-x 1 ubuntu ubuntu 97 Jun  1 04:20 variables.sh
ubuntu@ip-172-31-44-9:~$ █

ubuntu@ip-172-31-44-9:~$ ./variables.sh
Hello, I am Parthavi and my role is Cloud Engineer
ubuntu@ip-172-31-44-9:~$ █

```

1. Single Quotes '...'

Preserve the literal value of all characters inside the quotes.

Variables are not expanded or interpreted.

Special characters are treated as plain text.

Example:

```

NAME="Parthavi"
echo 'Hello, I am $NAME'

```

Output:

```
Hello, I am $NAME
```

Note: \$NAME is printed literally, not replaced with the variable's value.

2. Double Quotes "..."

Allow variable expansion.

Variables inside double quotes are replaced with their current values.

You can include special characters that will be interpreted (like \n for newline).

Example:

```
NAME="Parthavi"
echo "Hello, I am $NAME"
```

Output:

Hello, I am Parthavi

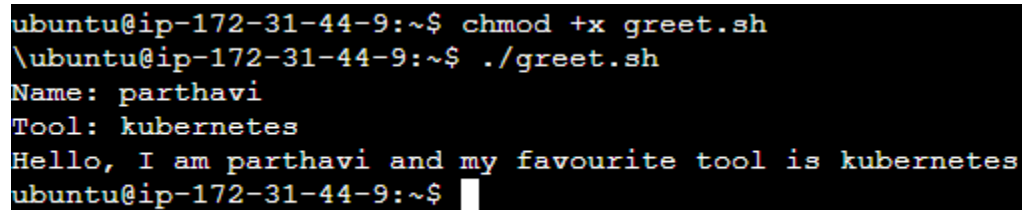
Note: \$NAME is replaced with the value of the variable.

Task 3: User Input with read

```
#!/bin/bash
```

```
read -p "Name: " name
read -p "Tool: " tool
```

```
echo "Hello, I am $name and my favourite tool is $tool"
```



```
ubuntu@ip-172-31-44-9:~$ chmod +x greet.sh
ubuntu@ip-172-31-44-9:~$ ./greet.sh
Name: parthavi
Tool: kubernetes
Hello, I am parthavi and my favourite tool is kubernetes
ubuntu@ip-172-31-44-9:~$
```

Task 4: If-Else Conditions

```
#!/bin/bash
```

```
read -p "Enter Number: " number
```

```
if [[ $number -gt 0 ]]; then
    echo "$number is positive."
elif [[ $number -lt 0 ]]; then
    echo "$number is negative."
else
    echo "$number is zero."
fi
```

Output:

```
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: 3
3 is positive.
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: -9
-9 is negative.
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: 0
0 is zero.
```

```
ubuntu@ip-172-31-44-9:~$ vim check_number.sh
ubuntu@ip-172-31-44-9:~$ chmod +x check_number.sh
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: 3
3 is positive.
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: -9
-9 is negative.
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: 0
0 is zero.
ubuntu@ip-172-31-44-9:~$
```

file_check.sh

File_check.sh

#!/bin/bash

read -p "Enter the filename: " filename

```
if [[ -f "$filename" ]]; then
    echo "The file '$filename' exists."
else
    echo "The file '$filename' does not exist."
fi
```

Output:

```
ubuntu@ip-172-31-44-9:~$ ./file_check.sh
Enter the filename: notes.txt
The file 'notes.txt' exists.
ubuntu@ip-172-31-44-9:~$ ./file_check.sh
Enter the filename: file.txt
The file 'file.txt' does not exist.
ubuntu@ip-172-31-44-9:~$ nano file_check.sh
```

```
ubuntu@ip-172-31-44-9:~$ ./file_check.sh
Enter the filename: notes.txt
The file 'notes.txt' exists.
ubuntu@ip-172-31-44-9:~$ ./file_check.sh
Enter the filename: file.txt
The file 'file.txt' does not exist.
ubuntu@ip-172-31-44-9:~$
```

Task 5: Combine It All

Server_check.sh

```
#!/bin/bash
```

```
# Store service name in a variable
service="sshd"
```

```
# Ask user for confirmation
read -p "Do you want to check the status? (y/n): " choice
```

```
if [ "$choice" = "y" ]; then
    if systemctl is-active --quiet "$service"; then
        echo "$service is active."
    else
        echo "$service is not active."
    fi
fi
```

```
# Show detailed status
systemctl status "$service"
elif [ "$choice" = "n" ]; then
    echo "Skipped."
else
    echo "Invalid input. Please enter y or n."
fi
```

Output:

Do you want to check the status? (y/n): y

sshd is active.

or

Do you want to check the status? (y/n): n

Skipped.

```
ubuntu@ip-172-31-44-9:~$ ./server_check.sh
Do you want to check the status? (y/n): y
sshd is active.
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Drop-In: /usr/lib/systemd/system/ssh.service.d
            └─ec2-instance-connect.conf
   Active: active (running) since Tue 2026-06-02 03:33:23 UTC; 6min ago
   Invocation: d5528ee815024508b7a8699ddfe254a4
   TriggeredBy: ● ssh.socket
   Docs: man:ssh(8).
          man:ssh_config(5).
   Process: 623 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 657 (sshd)
   Tasks: 1 (limit: 627)
   Memory: 7.8M (peak: 14.8M)
   CPU: 1.134s
   CGroup: /system.slice/ssh.service
```

```
ubuntu@ip-172-31-44-9:~$ ./server_check.sh
Do you want to check the status? (y/n): n
Skipped.
ubuntu@ip-172-31-44-9:~$
```

